A BM EDITOR MANUAL

bm Studio: 3D modelling

Build 3D models for your game on the console, from tiles and blocks, and use them in your code.

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README

What this is. The manual of bm Studio, the console program that makes the 3D models of a bm game. It is written from the program's own code and docs.

How to use it. Read the first three chapters once, then follow the tutorials with the console in your hands. Anything in a bordered box with a shadow is meant to be copied as it is.

What is missing. Every gap is marked with a highlighted TO FILL tag; nothing is invented.

What bm Studio is

bm Studio builds 3D models out of blocks and tiles, right on the console. You place a cube, stick a piece of your sprite sheet on its walls, move a corner to make a roof, and the model is ready to be drawn by your game with `draw3d()`.

What it connects to

- **The project.** bm Studio reads and writes the MESH section of your project (a `.bme` file). It does not touch the game code, the map or the sections it does not know.
- **bm Animator.** Skeletons and animations are made there. In bm Studio's menu, *Open in bm Animator* opens the same file in it.
- **bm Mesh.** The analytic editor of vertices and triangles, including the meshes your code builds. It works on the same file.
- **bm Pixel and the sprite sheet.** The textures of the faces are pieces of the project's sprite sheet. bm Pixel draws it.
- **bm SDK.** The hub of the project. In its F1 page the *OPEN IN* list has bm Studio as number 3, and the menu of bm Studio has *Back to bm SDK* when the SDK opened it.

Which editor for what

- **bm Studio**: static models. Objects, props and buildings, made of blocks and tiles. The 3D map of a game is made in the map page of bm SDK.
- **bm Mesh**: organic models. Characters and heroes, and all fine editing of vertices and triangles.
- **bm Animator**: rigs and animations. It puts a skeleton into a model and moves it. It does not build models.
- bm Studio and bm Mesh write the same MESH section, so a model can go from one editor to the other.

Good to know

- bm Studio is a cartridge built into the kernel. You find it in the **Dev** tab of the menu.
- You use it with the keyboard or the gamepad (Bluetooth or USB). The mouse also works when one is connected.
- Hold **F12**, or press **?**, and the keys of the page you are on appear on the screen.

Opening a project or a game

bm Studio changes **projects** (`.bme`). A **game** (`.bm` or `.b16`) is only read: you can open it to look at its models, and the first save makes an editable copy.

Ways to start

From	What you do
The menu	Open bm Studio in the **Dev** tab
A game in the menu	**X** on its cover, then *Open in bm Studio*
bm SDK	F1 page, *OPEN IN*, number 3
bm Animator	In its menu, *Open in bm Studio*

Inside bm Studio

- Press **Esc** to reach the menu. *Open...* lists the `.bme` projects of `/bme` and the `.bm` and `.bme` files of `/carts` and of the root of the SD card. Shortcut: **Ctrl+O**.
- *Import a model...* (**Ctrl+I**) and *Export to /bmm...* (**Ctrl+E**) bring a model in from `/bmm` or `/glb` and write one to `/bmm` (see "Saving and exporting").
- *New project* (**Ctrl+N**) starts an empty project. It comes with the starter tiles of bm Studio on its sprite sheet.
- A new project is not saved yet: the menu shows `(not saved yet)` next to the name. Save it with **Ctrl+S**.

Caveats

- Opening a game says `a game: saving makes its copy` . The first save asks `Save an editable copy: name.bme?` ; from then on you work on the copy.
- A game without models can receive some: build them and save.
- If the menu shows `*modified*` , there are changes not saved. *Open...*, *New project* and *Exit* ask before throwing them away.

The screen and the 3D view

bm Studio has two pages, chosen with the F keys (or **Y + left/right** on the gamepad): **F1 build** and **F2 models**. **Esc** (on the gamepad **Y + B**) opens the menu.

The build page

- On the left, a panel: the five tools, the brush (a tile or a colour, with the sheet around it) and the model (faces, triangles, vertices).
- Over the view, two lines say what is there and what is being done.
- At the bottom, the keys of the moment.
- The cursor is a cell-shaped box. The grid on the floor is one unit per square.

[TO FILL: shots/editor-studio-build.png]

The 3D view controls

Key	Gamepad	What it does
`q` / `e`	X + left / right	Turn the view by 45 degrees
`w` / `s`	-	Tilt the view
`Alt` + left / right	-	Turn the view by 45 degrees (build page)
`Alt` + up / down	-	Tilt the view
`+` / `-`	X + A / X + B in the models page; none on the build page	Zoom in / out
`z`, `0` or `Home`	-	Frame the whole model
`v`	-	Change the look: light, flat, wireframe
`b`	-	Show the faces seen from behind, darker
`g`	-	Show or hide the grid (block and tile tools)
`F12` or `?` held	-	Show the keys of the page

With a mouse: in the view, the right button dragged turns and tilts, the middle button dragged moves, the wheel zooms. The pointer is the cursor: the left button does what space does.

Good to know

- Moving the cursor is always relative to the view. After you turn the view, up still means "away from you".
- The status line shows `model 1/3: house` : the model you are on, out of how many.
- Press **[** or **]** to go to the previous or next model from any page.
- **Ctrl+Z** undoes, **Ctrl+Y** redoes.

Building: tiles and blocks

The first two tools put things in a cell. Choose them with **1** (block) and **2** (tile).

The block tool

A block is a cube of one unit. Two blocks side by side have no wall between them, and removing one brings back the neighbour's wall.

Key	What it does
arrows	Move the cursor on the floor plane
`PgUp` / `PgDn`	Move the cursor one level up / down
`Space`	Put a block
`Backspace` or `Del`	Remove the block
`k`	Change the kind of block: cube, ramp, corner, rounded

On the gamepad: **A** puts, **B** removes, **X + up/down** changes level.

Ramp, corner and rounded blocks

Press **k** (with the block or tile tool) to change the kind of block the next **Space** puts: cube, ramp, corner, rounded. The panel names the kind when it is not the cube. The kind stays until you change it.

Kind	Shape	Triangles
cube	the cube of one unit	12
ramp	a slope from the near floor edge up to the far top edge, a wall on the far side, two triangular sides	8
corner	a pyramid on the floor of the cell, its top at the far right top corner; its sloped sides fit a ramp	6
rounded	an eight-sided column: the four vertical edges are cut 0.3 deep	28

The ramp and the corner rise **away from the camera**. Turn the view with **q** and **e** (45 degrees) to turn the block: far and right follow the view, in steps of 90 degrees. Every face takes the brush (a colour or a tile). **Backspace** removes the block of the cell, of any kind, and undo takes it back. The walls of a neighbour that look into the cell stay.

The tile tool

A tile goes on one side of the cell, not on a whole cube. Use it for floors, single walls, doors and windows.

- Press **f** to choose the side in turn: floor, far wall, right wall, near wall, left wall, ceiling. On the gamepad, tap **X**.
- **Space** puts the tile, **Backspace** removes what is on that side.

The brush: tiles and colours

Press **Tab** (on the gamepad **Y**) to open the sheet.

Key	What it does
arrows	Move over the sheet
`z`	Tile size: 8, 16 or 32 pixels
`a` / `d`	Narrower / wider selection
`w` / `s`	Shorter / taller selection
`c`	Switch between tiles and the palette of colours
`Enter`	Take it as the brush
`Tab`	Close without changing

- A selection of several tiles is placed in one go: a door two tiles high is one put.
- Outside the sheet, **r** turns the brush by a quarter turn with tools 1-2 only (with tool 3, select, **r** turns the chosen faces), and **h** mirrors the brush with any tool. **x** picks the tile or colour from the face under the cursor.
- With a colour as brush, the faces get a flat colour and no texture.

[TO FILL: shots/editor-studio-tiles.png]

Selecting and moving

Select (tool 3)

The arrows bring the pointer to the nearest face in that direction on screen.

Key	What it does
`Space`	Choose or release the face
`a`	All faces, or none if some are chosen
`c`	All the faces joined to the chosen ones: a whole object
`g`	Move (see below)
`r` / `t` / `m`	Turn / flip / mirror
`n`	Show the other side of the face
`Enter`	Put the brush's tile on the faces
`u`	Turn the texture
`,` / `.`	Half as big / twice as big
`d`	Copy and move the copy
`Del`	Delete the faces
`o`	Move the faces into a new model

Move (g)

Arrows move, `PgUp` / `PgDn` go up and down. **Tab** changes the step, from 1 down to 1/16. **Enter** leaves the faces there; **Esc** puts them back where they were.

Vertex (tool 4)

The pointer goes on the corners. **Space** chooses, **a** all or none, **g** moves them (roofs, ramps, free shapes), **m** joins several corners into one.

Painting a model

bm Studio paints **pixels of a face's tile**, not colours at the corners. There is no vertex colour: a face is either a tile of the sheet or one flat colour.

What it does

With tool **5** (paint) chosen, **Enter** or **Space** opens the tile of the face under the pointer, large in the panel. You paint it pixel by pixel while the model on the view changes. The painted sheet is saved with the model.

Good to know

- The painted pixels are on the project's sprite sheet. A tile used by several faces changes on all of them.
- A face with a flat colour is not painted pixel by pixel: **Enter** gives it the paint colour in one go.
- The sheet is the same one bm Pixel and your game use. If your game changes it with `sset`, the models change too.

Key	What it does
arrows	Move the pixel cursor
`Space`	Paint the pixel
`e`	Clear the pixel (transparent)
`i`	Pick the colour of the pixel under the cursor
`c`	Choose a colour: the palette, then the colours already in the tile
`Esc` or `Enter`	Close the paint panel

With a mouse, hold the left button over the pixels and drag.

[TO FILL: shots/editor-studio-paint.png]

The models page and mesh reduction

Press **F2**. The list on the left holds the models of the file. Over the view, three lines show the name, the counts (faces, triangles, vertices) and either the skeleton (`N bones, N animations`) or `no skeleton (bm Animator makes one)` . A blue dot marks a model with a skeleton.

Keys

Key	What it does
up / down	Choose the model
`Enter`	Build it (go to the build page)
`n`	New model
`r`	Rename (up to 16 letters)
`d`	Duplicate
`Del`	Delete (press twice)
`PgUp` / `PgDn`	Change the order
`-`	Reduce the triangles
`m`	Make a model from a picture
`i`	Texture margin: 0, 0.125, 0.25, 0.5 pixels

[TO FILL: shots/editor-studio-models.png]

Reducing triangles

Press `-` . The box asks for the triangles wanted and proposes half. The reducer keeps borders, colour lines and texture seams, and the skeleton follows the vertices. **Ctrl+Z** undoes it.

Why it matters

The console draws about 1200 triangles in a scene at 60 fps. Beyond that, the counts turn to a warning: `heavy for 60 fps (1200 triangles a scene)` .

Caveats

- If you ask for more triangles than the model has, you read `it has N triangles already` .
- Sometimes the model stays above the number asked: the message adds `(no further without turning faces over)` .
- On the models page the key `-` reduces, so zoom with `+` or the mouse wheel.

A model from a picture

`m` (or *Model from picture...* in the menu) makes a model from an image in the `pics/` folder of the SD or in the SD root (`.png` or `.jpg`). There are three ways: **cutout** (a thick paper figure), **lathe** (the half outline turned around the vertical axis: vases, towers) and **meshy.ai** (an online service that needs the network and an account).

Saving and exporting

Key	What it does
`Ctrl+S`	Save in place
`Ctrl+Shift+S`	Save as: a name in 8.3 form, in `/bme`, ending in `.BME`
`F5`	Save, then try the game

What is saved

- The models go in the MESH section of the file, together with the sprite sheet if you painted it. The skeletons of the models are saved with them.
- The menu lets you set the cartridge's title and author.
- A new project is born with a small viewer program: it shows the models, so F5 works at once.

Resource files

The resource format for models is `.bmm`: MESH, ANIM when animated and a sheet with only the cells the textures use. On the PC, `scripts/bmres.py extract GAME.bm --model villager -o VILLAGER.bmm` makes one from a game.

bm Studio reads and writes `.bmm` files in `/bmm`. The next section shows how.

Import a .glb

The SD card has two folders for models. `/glb` holds `.glb` files that you copy there; bm Studio only reads them. `/bmm` holds the converted models (`.bmm`); bm Studio writes them there. All the names are in lower case.

1. Press **Ctrl+I**, or choose **Import a model...** in the menu (Esc). On the pad, press Y + B for the menu.
2. The list shows `/bmm` and `/glb`. A name marked `.glb: converted to /bmm` has no `.bmm` yet.
3. Choose a `.bmm` to read it. Choose a `.glb` to convert it: bm Studio writes `/bmm/NAME.bmm` and adds the model to the project.
4. If the name has more than 8 characters, or `/bmm` has the name already, bm Studio asks for a new name (1 to 8 of a-z, 0-9, `_` and `-`).
5. The status line shows the vertices and the triangles. If the file is not good, it shows the message of the console, for example the format limit of 65535 vertices or triangles.

A `.glb` gives flat colours. A `.bmm` with a skeleton keeps it (bm Animator plays it). The texture of a `.bmm` goes on the sheet of the project only when the sheet is empty; if not, the faces that used it turn grey. A `.bmm` with several models gives the first one.

Ctrl+S saves the model in the project, as for any model. **Ctrl+E**, or **Export to /bmm...** in the menu, writes the current model to `/bmm/NAME.bmm` with its skeleton; if the file is there, bm Studio asks before it replaces it.

For a game built from the repository

A `.bm` with the models and skeletons, saved by bm Studio, can be put as `models.bm` in `carts/<game>/`. `make` puts it into the cartridge with `mkbm.py --models`. If the game has no `sheet.png`, the sheet is the one of `models.bm`.

Using the model in a game

The functions below are documented in the 3D API.

Function	What it gives
`models()`	The names of the cartridge's models, in order
`model(name)`	The model as a mesh (`nil` if missing). Call it in `_init`
`bounds3d(m)`	`x0, y0, z0, x1, y1, z1` , the box around the mesh
`draw3d(m, x, y, z, rx, ry, rz, scale, flags)`	Draws the mesh
`camera3d` , `light3d` , `zclear`	Camera, light, clear the depth

COPY AND PASTE

```
local casa
function _init()
  casa = model("house")
end
function _draw()
  cls(0x1c2030)
  zclear()
  camera3d(0, 4, -8, 0, -0.4)
  light3d(-0.4, 0.8, -0.5, 0.4)
  draw3d(casa, 0, 0, 0, 0, time() * 0.5)
end
```

Good to know

- The grid origin of bm Studio goes at the point `x, y, z` of `draw3d` . One square is one unit.
- Textured faces use the sprite sheet of the cartridge.
- Models with a skeleton are animated with `animate()` ; see the bm Animator manual.

Guided tutorials

Your first block

1. Open bm Studio from the **Dev** tab.
2. Press **Esc**, choose *New project*. You see the build page with tool **1 block** chosen.
3. Press **Space**. A cube appears in the view.
4. Press **PgUp**, then **Space**. A second cube sits on top of the first.
5. Press **q** a few times. The view turns by 45 degrees each time.
6. Press **Backspace**. The block under the cursor disappears.
7. If you made a mistake, press **Ctrl+Z**.

> **TRY IT**
> Put three blocks in a row. Check in the panel that `faces` and `tri` count only the visible walls.

Paint a model

1. Put one block with tool **1**.
2. Press **5** to choose paint, then **Enter**. The tile of the face under the pointer appears large in the panel.
3. Press **c** and choose a colour from the list.
4. Move with the arrows and press **Space**: the pixel takes the colour, and so does the face on the view.
5. Press **e** on a pixel to clear it.
6. Press **Esc** to close the panel and **Ctrl+S** to save.

If the face has a flat colour instead of a tile, **Enter** paints the whole face. To paint pixels, first give the face a tile: tool **3**, choose the face with **Space**, press **Tab**, pick a tile and press **Enter**, then **Enter** again to put it on the face.

Reduce a mesh and see the triangle count

1. Press **F2**. Choose a model with many triangles.
2. Read the second line over the view: the faces, the `tri` and the vertices.
3. Press `-` . The box proposes half of the triangles. Press **Enter**.
4. You read `reduced to N triangles (Ctrl+Z undoes)` and the count changes.
5. Press **Ctrl+Z** to go back.

> **BEWARE**
> If the count is above 1200, the line says `heavy for 60 fps` . Reduce until the warning goes away.

Put your model in a game

1. Build and name a model, for example `house` . On the models page, **r** renames it.
2. Press **Ctrl+S**. A new project asks for a name: type one in 8.3 form.

3. Press **F5**. The game runs with the viewer that bm Studio put in the project; left and right change the model.
4. Open the same project in bm Code (bm SDK, F1 page, *OPEN IN*, number 1) and replace the viewer's code with the example of the previous chapter, using your model's name in `model("house")`.
5. Press **F5** again. You see your model turning.

Tips and limits

- About 1200 triangles in a scene is what the console draws at 60 fps. A model you reduced is the cheapest way to stay under it.
- Cursor cells go from -64 to 63 on each axis.
- The panel counts vertices and triangles. There is no size cap on saving; only the format limit of 65535 vertices and 65535 triangles per mesh stays. The vertex counter shows it ("of 65535").
- Textured faces need the sprite sheet: the textures are pieces of it.
- Use **F5** often: it saves and shows the model in the game.
- A game and a project are not the same: the menu shows the path of the file you are working on, so check it is the `.BME` copy before you go on.
- **[TO FILL: the gamepad keys of bm Studio tried on the PowKiddy RGB30]**

Open points

- **The screen and the 3D view**: picture to take: shots/editor-studio-build.png
- **Building: tiles and blocks**: picture to take: shots/editor-studio-tiles.png
- **Painting a model**: picture to take: shots/editor-studio-paint.png
- **The models page and mesh reduction**: picture to take: shots/editor-studio-models.png
- **Tips and limits**: the gamepad keys of bm Studio tried on the PowKiddy RGB30